

Sentiment and Hate Analysis of Mpox Instagram Dataset

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Abstract

The advent of social media has reshaped how we can assess public sentiment, with user-generated content providing a rich source of information on individual and collective views. This report explores the Mpox Instagram dataset, which contains posts with sentiment annotations, to develop machine learning models for the classification of sentiment. By employing natural language processing (NLP) techniques and algorithms such as Decision Trees, Logistic Regression, and Random Forest, this study aims to uncover patterns in both the captions and images of posts related to Monkeypox. The research methodology includes data preprocessing, feature extraction, and model evaluation. Initially, the data is cleaned to remove any inaccuracies, ensuring it is suitable for analysis. Feature extraction then converts the raw text into a format that can be processed by machine learning models. Following the preparation of the data, the models are trained and their effectiveness evaluated using key metrics, such as accuracy and confusion matrices. The findings indicate that these machine learning models are effective for sentiment classification tasks. The Decision Tree model, in particular, is noted for its interpretability, making it an ideal choice when transparency is required for understanding the model's decisions. The evaluation metrics, including accuracy and confusion matrices, offer a comprehensive overview of the models' performance, highlighting both their advantages and areas that could benefit from further development.

1 Introduction

Social media has revolutionized digital-age communication. Platforms like Instagram host a continuous flow of opinions, reactions, and personal viewpoints that mold online conversations. By examining these emotional expressions in Instagram posts, researchers can uncover valuable insights into how different communities perceive and re-

spond to various issues. This understanding aids organizations and leaders in comprehending what drives public sentiment and engagement. Analyzing emotional content in text necessitates advanced computational methods. The proposed research employs state-of-the-art sentiment analysis techniques to process user-generated content from Instagram. Natural Language Processing (NLP) frameworks and machine learning systems can be utilized to systematically analyze large text datasets to detect emotional signals—whether positive, negative, or neutral. The current study focuses on emotional patterns in Instagram discussions about Mpox, using a dataset where posts are labeled with sentiment indicators. The research aims to develop intelligent systems capable of automatically identifying and categorizing emotional tones in Instagram content. A multi-model approach has been created using sentiment labels as classification targets, that integrates several machine learning algorithms. The framework combines specialized implementations of Decision Trees, Logistic Regression, and Random Forest algorithms to build a robust classification system that maintains accuracy even with large data volumes. To ensure reliable outcomes, a three-phase analytical framework has been established. Initially, thorough data cleaning protocols have been implemented to standardize the dataset. Next, advanced feature engineering techniques have been applied to convert raw text into structured numerical representations suitable for machine learning analysis. Finally, comprehensive performance evaluations have been conducted using multiple metrics, including precision measurements and detailed confusion matrices. The Decision Tree component is particularly valuable due to its transparent architecture, that allows to trace exactly how the system arrives at its emotional classifications. While the current research focuses on Mpox-related content, the proposed analytical framework is designed to be adaptable to various topics and contexts. This

flexibility ensures that the proposed methods can be applied to understand emotional patterns in different social media discussions, providing a foundation for future sentiment analysis applications. The technical approach prioritizes reproducibility and scalability, ensuring that the proposed methods can be effectively applied to the ever-growing volume of social media content. By developing these specialized analytical tools, the field's ability can be enhanced to understand and interpret the emotional dimensions of online communication.

2 Dataset Description

The Mpox Instagram Dataset forms the cornerstone of this project, designed to facilitate sentiment and hate speech analysis in social media contexts. Sourced from Kaggle, this dataset focuses on Instagram posts discussing the Mpox outbreak, making it a valuable tool for examining how social media conversations reflect and influence public perceptions during health crises.

The dataset includes several essential components that are critical for analysis:

1. Post ID: A unique number assigned to each post for seamless identification and tracking throughout the analysis process.

2. Translated Post Description: The text content of each post, translated into English, capturing the expressions and emotions shared by users. This serves as the primary input for sentiment analysis.

3. Sentiment: A label indicating the emotional tone of each post, categorized into "Positive," "Negative," or "Neutral." This variable acts as the target for training machine learning models.

4. Hate Speech Label: A binary attribute that identifies whether a post contains hate speech, classified as either "Yes" or "No." This is crucial for detecting and analyzing harmful language in online interactions.

The dataset is characterized by its diversity, featuring a wide range of sentiments and viewpoints. This variation is invaluable for training machine learning algorithms, as it exposes the models to a broad spectrum of linguistic styles and emotional expressions. The ample size of the dataset ensures its effectiveness for both training and validation tasks, enhancing the reliability of sentiment classification models.

Through the analysis of this dataset, the project aims to identify patterns in public sentiment and detect instances of hate speech during a global health

crisis. The findings will provide insights into how social media discourse can impact public health strategies, reveal areas requiring targeted communication, and suggest approaches to mitigating harmful content. This analysis contributes to understanding the role of online platforms in shaping opinions and addressing challenges in public health.

3 Research

The research component of this project focusses on the role of text preprocessing and feature extraction in boosting machine learning model performance for natural language processing (NLP) applications. These steps are essential for converting raw text into a structured format that machine learning models can easily analyse. Proper preprocessing and feature extraction lay the groundwork for accurate sentiment classification, enabling algorithms to interpret data rapidly.

The preprocessing phase began with case normalisation, which required transforming every text to lowercase in order to standardise the dataset. This stage reduced complexity by ensuring consistency across variations like "Mpox" and "mpox," allowing the model to identify patterns more effectively. Punctuation marks were also removed to accent the text themselves, removing nonverbal cues that could produce unwanted noise. These early methods are in line with established best practices in text preparation (Dahab et al., 2015).

Eliminating stop words was another critical step. To minimise noise, the Natural Language Toolkit (NLTK) was used to filter out common phrases such as "and" and "the". It is important to remember that some stop words may have emotional overtones, so extra care was taken to ensure that only relevant stop words were chosen. Lemmatisation was then employed to reduce words to their most fundamental forms, such as converting "running" to "run." By lowering dimensionality and consolidating word variants, this strategy helped to focus the research on the words' key meanings (Dahab et al., 2015).

Following preprocessing, feature extraction was used to transform the cleaned text into numerical data for machine learning models. Initially, Count Vectorisation was used to generate a document-term matrix, with each row representing an individual post and the columns representing unique terms. The numbers in the matrix represented the frequency of words in the posts, giving a numerical representation of the text data.

Three methods were considered when selecting a model: Random Forest, Logistic Regression, and Decision Trees. Because they provide clear insights into the decision-making process, decision trees were chosen for interpretability. They are, nevertheless, prone to overfitting, especially when working with complex datasets (Dahab et al., 2015). Logistic regression was chosen because to its competence in binary classification problems and ability to generate probabilistic outcomes. Random Forest, an ensemble strategy that combines numerous decision trees, helped to reduce overfitting and improve the model's predicted accuracy and robustness (Zhang et al., 2018).

The evaluation of model performance was a key part of this study. Accuracy was the most important measure, and confusion matrices were built to display classification results and identify areas where misclassification occurred. Word clouds were also created to highlight the most frequently used terms in each sentiment category, providing further context for the language patterns. The distribution of predicted classes was examined to ensure balance across the dataset, ensuring that the model was not biased towards any one sentiment. This rigorous evaluation technique offered a thorough grasp of each model's strengths and weaknesses, guiding the choice of the optimal approach for sentiment categorization in the context of social media data (Kumar et al., 2023), (Kaur et al., 2017).

4 Methodology

The methodology for this project is structured into several key phases, each designed to ensure a systematic and thorough approach to data analysis, model development, and evaluation. Below is a detailed outline of the steps undertaken throughout the project:

4.1 Data Loading

The first step in the methodology involved loading the Mpox Instagram dataset, which was stored in an Excel file, into a Pandas DataFrame. This was accomplished using the Pandas library's `read_excel()` function, which streamlines the process of importing data from Excel files into a structured format appropriate for analysis. The DataFrame format enabled easy data manipulation and exploration. The dataset was evaluated by presenting the first few rows, which gave an overview of the data structure and content, including post descriptions and

sentiment labels. To better comprehend the dataset, the `df.info()` method was used to collect general DataFrame information, such as column data types and the occurrence of missing values. Descriptive statistics were also calculated for the numeric columns to examine the distribution of numerical features and find any anomalies or outliers.

4.2 Data Exploration

After loading the data, an exploratory data analysis (EDA) was performed to learn more about the dataset's structure and properties. The distribution of the desired trait, sentiment, was a major focus. This analysis was useful in determining class balance since it offered an overview of the frequency of each sentiment category in the sample. Understanding the distribution of sentiment labels is critical for ensuring that the model is trained on a representative sample, which can have a substantial impact on its performance and generalisation.

4.3 Text Preprocessing

Text preprocessing is an important step in preparing data for machine learning since it cleans and transforms text data to increase the quality of input characteristics. To do this, a custom transformer class was created to encapsulate the preprocessing functionality while adhering to the scikit-learn pipeline layout. The preprocessing steps used in this class are as follows:

1. **Lowercasing:** All text data was changed to lowercase to promote uniformity and simplicity. This phase helps to avoid differences caused by case sensitivity, ensuring that terms are treated consistently throughout the dataset.
2. **Punctuation Removal:** Punctuation marks were deleted with the `str.translate()` function and `string.punctuation`. This ensured that punctuation did not interfere with the analysis, allowing the model to focus entirely on the text's substance.
3. **Stop Word Removal:** Common terms with minimal meaning (such as "and," "the," and "is") were filtered out using a predefined list from the NLTK library. This stage reduced background noise, allowing the algorithm to focus on more significant phrases.
4. **Tokenization and Lemmatization:** The text was tokenised, which separated it into distinct words. Lemmatisation was then used to reduce words to their base forms, guaranteeing that variants of a word (for example, "running" and "run")

were considered as the same object. This reduced dimensionality and enhanced the model's accuracy.

4.4 Feature and Target Separation

After the text data had been preprocessed, the dataset was prepared for modelling by separating the descriptive features (the Instagram post descriptions) from the target feature (the emotion label). This stage was critical for training the machine learning models, ensuring that they could learn from the input features and predict the appropriate sentiment. The descriptive features were assigned to the variable X, while the target sentiment labels were assigned to y, in accordance with typical machine learning norms for feature and target variable representations.

4.5 Data Partitioning

To create strong machine learning models, the dataset was divided into training and validation sets using scikit-learn's `train_test_split()` tool. This function allows for the random division of the dataset into two subsets, one for training and the other for evaluating model performance. The partitioning technique incorporated stratification, which guaranteed that the distribution of sentiment labels remained consistent throughout both the training and validation sets. Stratification was critical to ensuring that the model was exposed to a balanced representation of each class, hence avoiding bias during training.

4.6 Model Training

To streamline the preprocessing and modelling stages, a machine learning pipeline was set up. This pipeline combined the proprietary text preprocessing transformer, feature extraction using Count Vectorisation, and the classification model. The model training phase consisted of the following steps:

1. Pipeline Creation: The pipeline was built using scikit-learn's Pipeline class, which allows many processing steps to be executed in sequence. The pipeline includes text cleaning, turning the text to a token count matrix, and classifying the results. This modular approach ensured that each step was completed in the proper order.

2. Model Selection: The first model used for training was a Decision Tree, especially the `DecisionTreeClassifier` from scikit-learn. Decision Trees were chosen because they are interpretable

and allow for easy visualisation of the decision-making process. The model was instantiated with the criterion set to 'entropy', which quantifies the information gain at each split and guides the tree's growth based on the purity of the nodes produced.

3. Training the Model: The model was trained on the training data using the specified workflow. This entailed fitting the model to the data and modifying its parameters based on the training set. The training phase was crucial because it allowed the model to understand the underlying patterns and relationships in the data.

4.7 Model Evaluation

To assess the trained model's performance, predictions were made using the validation dataset. The accuracy of the model was calculated using scikit-learn's `accuracy_score()` function, which compared the predicted labels to the actual labels in the validation dataset. This statistic offered an easy assessment of the model's performance. The Decision Tree algorithm attained an accuracy of 68%. In addition, a confusion matrix was created to provide a detailed analysis of the model's predictions versus the actual labels. This matrix helped identify true positives, true negatives, false positives, and false negatives, providing information on the model's classification performance. The `ConfusionMatrixDisplay` class was used to visualise the confusion matrix, which improved interpretability.

4.8 Alternative Models

Alternative algorithms were tested in order to increase model accuracy even further. A Random Forest classifier was created using scikit-learn's `RandomForestClassifier`. This ensemble approach combines multiple decision trees to improve resilience and prevent overfitting by averaging their predictions. The model was trained with the same pipeline structure and evaluated on the validation set, yielding an accuracy of 74%. Furthermore, Logistic Regression was utilised since it is effective in binary classification problems. The `LogisticRegression` class was created with a maximum iteration limit to assure convergence during training. This model was also added to the pipeline, and its performance was evaluated similarly to the Decision Tree and Random Forest models, with the greatest accuracy of 75%.

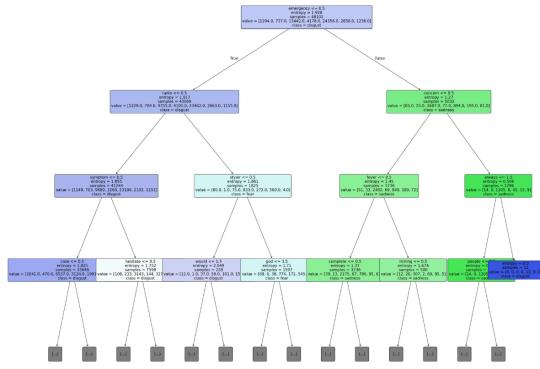


Figure 1: Decision Tree Using "Entropy" splitting criterion

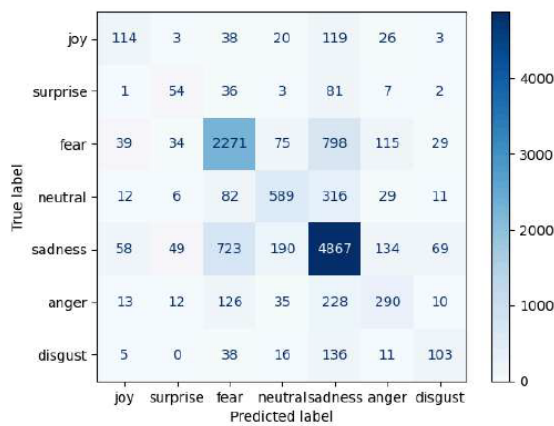


Figure 2: Confusion matrix

4.9 Visualization and Insights

Several visualisations were generated in order to acquire a better understanding of the text data and model performance. Each sentiment category was represented by a Word Cloud created with the WordCloud library. This visualisation highlighted the most commonly used words in the text data, making it easy to discover the dominating themes and phrases connected with each sentiment. The anticipated class distribution was further examined using Seaborn's countplot() method. This visualisation depicted the frequency of each predicted class, assisting in identifying any overfitting or underfitting concerns within the model.

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